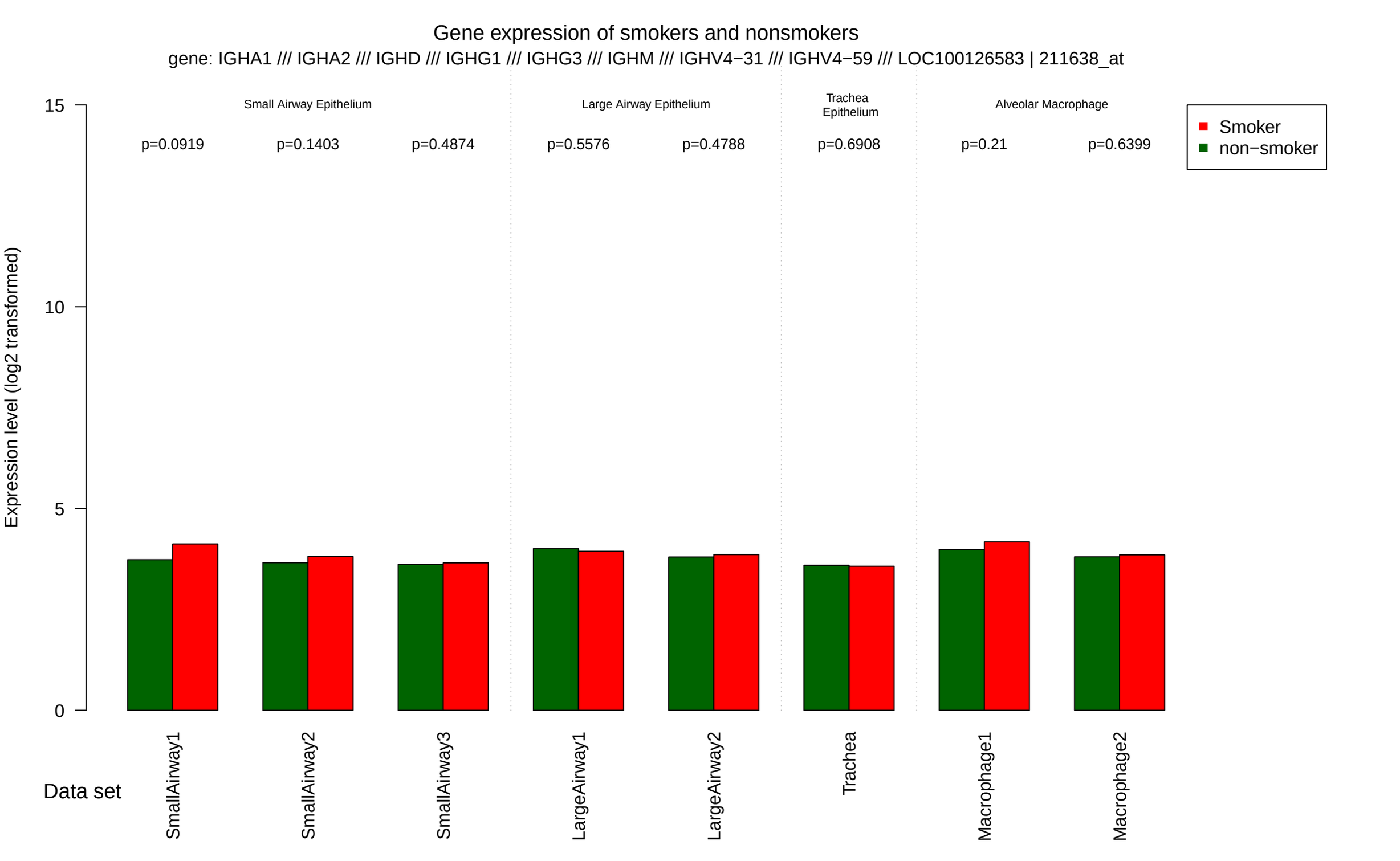
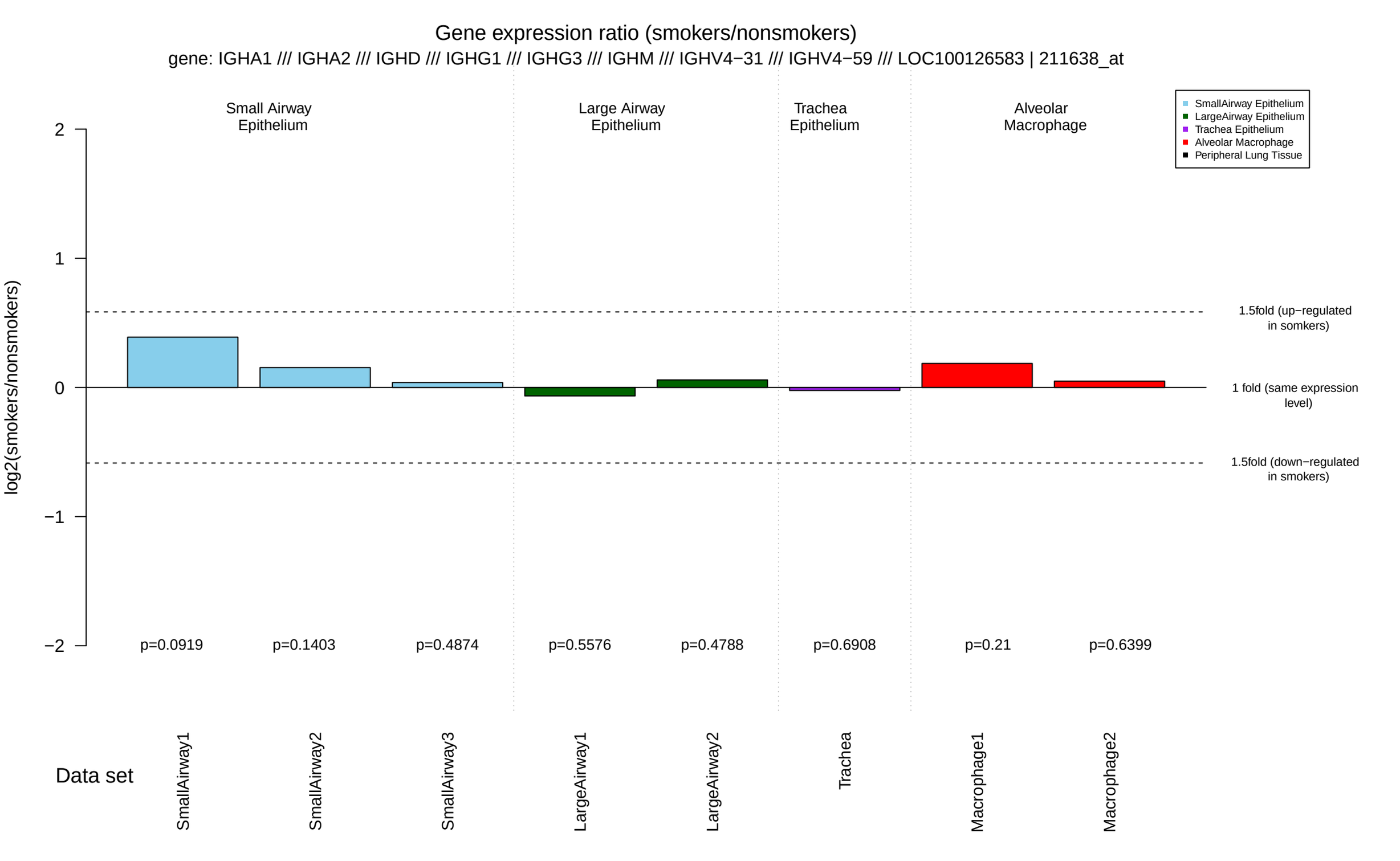
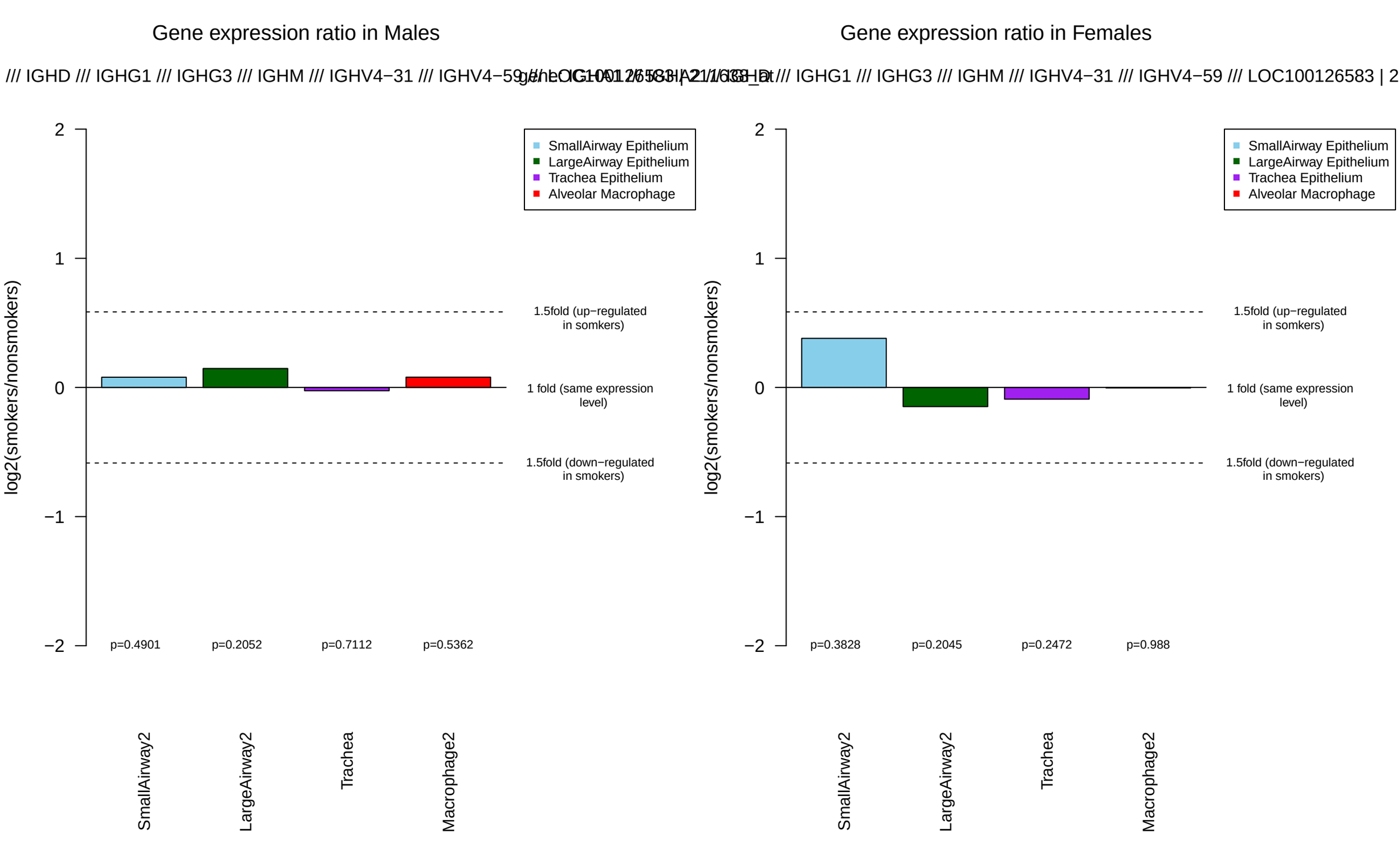
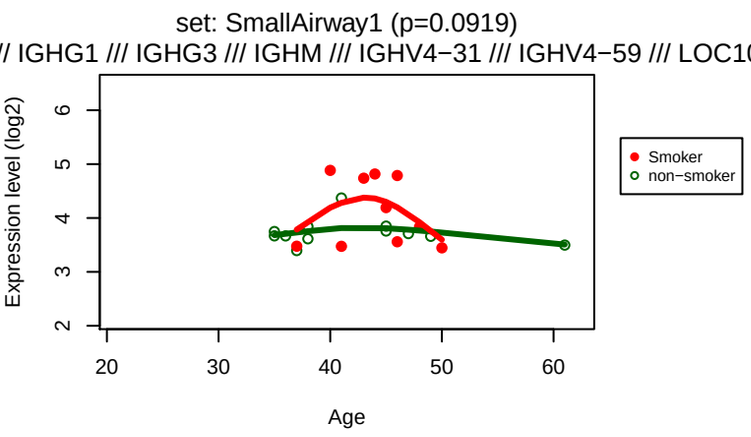
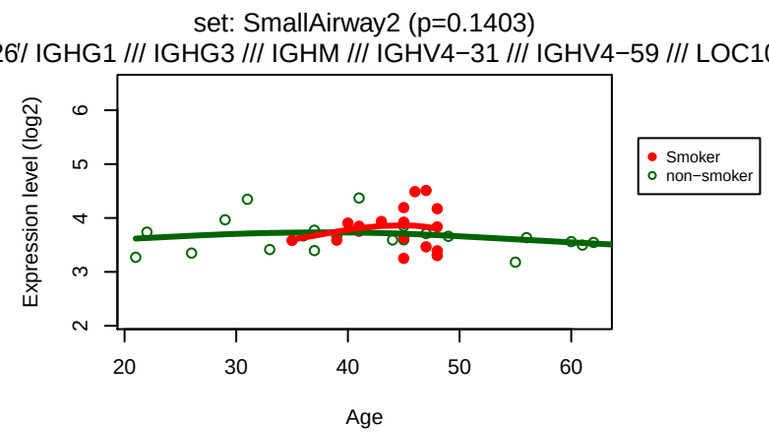
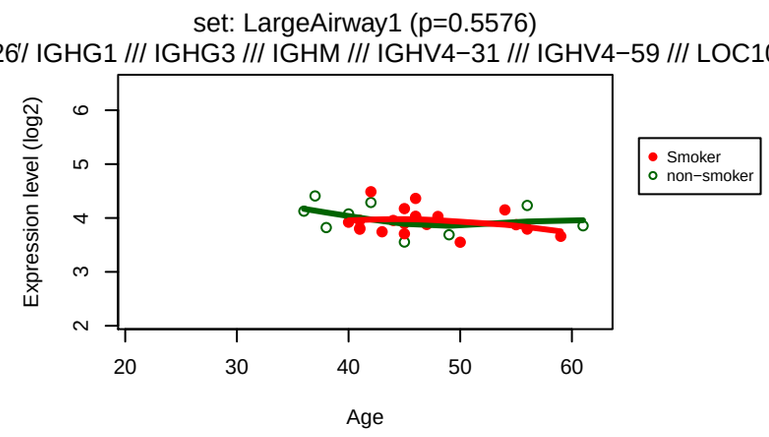

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211638_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0919
p=0.1403
p=0.4874
p=0.5576
p=0.4788
p=0.6908
p=0.21
p=0.6399
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

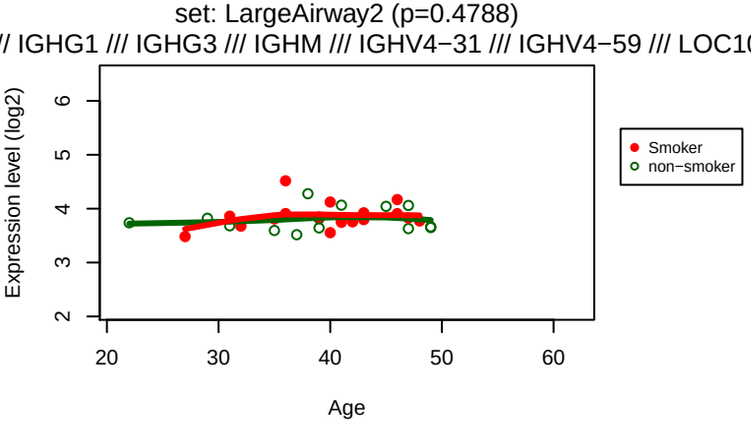

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211638_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0919
p=0.1403
p=0.4874
p=0.5576
p=0.4788
p=0.6908
p=0.21
p=0.6399
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

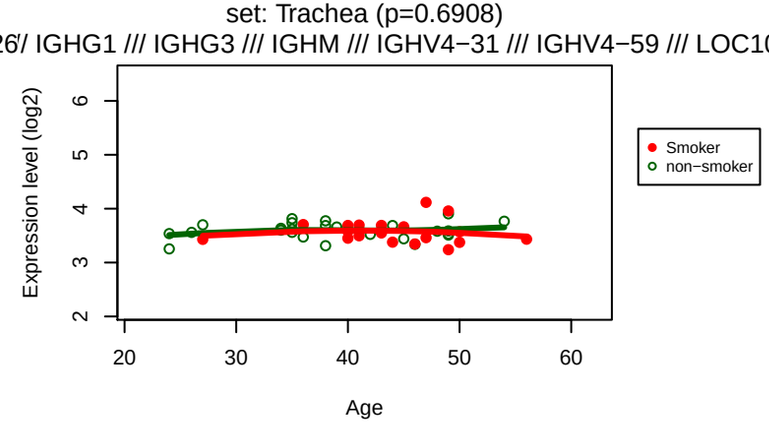

Gene expression ratio in Males
/// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 21638_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4901
p=0.2052
p=0.7112
p=0.5362
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
/// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 2
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3828
p=0.2045
p=0.2472
p=0.988
SmallAirway2
LargeAirway2
Trachea
Macrophage2

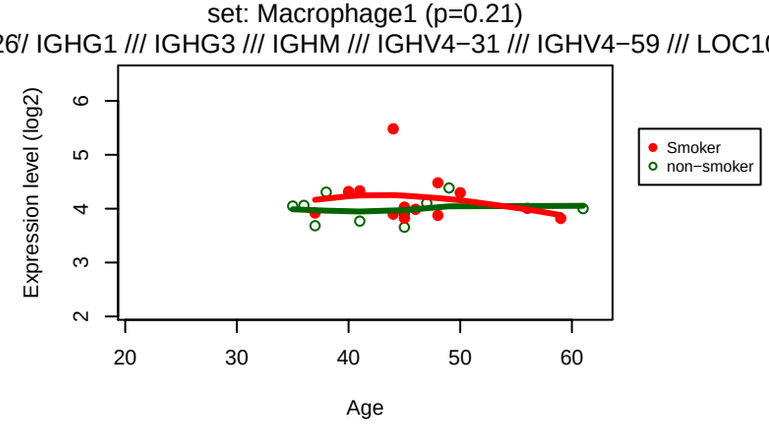

set: SmallAirway1 (p=0.0919)
/// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126
Expression level (log2)
Age
Smoker
non−smoker

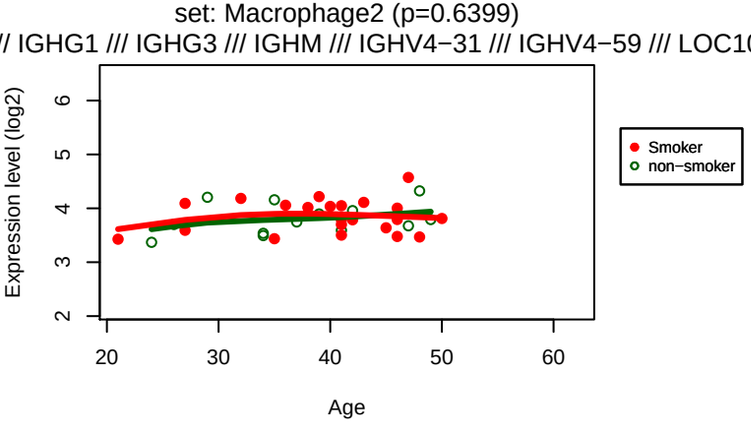

set: SmallAirway2 (p=0.1403)
/// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126
Expression level (log2)
Age
Smoker
non−smoker

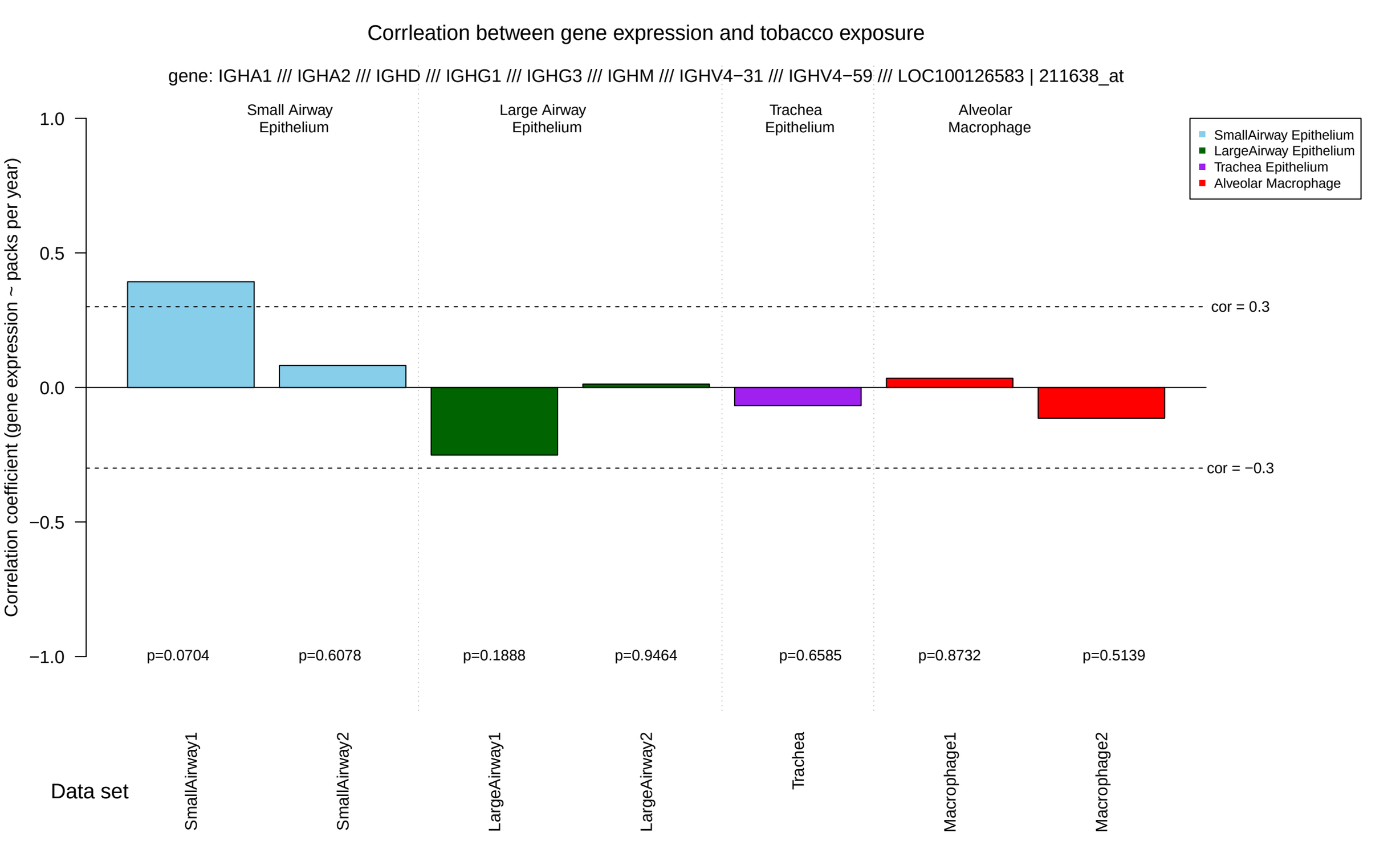

set: LargeAirway1 (p=0.5576)
/// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4788)
/// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6908)
/// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.21)
/// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6399)
/// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHM /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211638_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0704
p=0.6078
p=0.1888
p=0.9464
p=0.6585
p=0.8732
p=0.5139
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2